AI-POWERED ASSESSMENT PLATFORM

System Architecture, Module Responsibilities and Team Handover Plan

Project Objective

Design an integrated assessment platform that supports question-bank creation, syllabus-aware mock-test generation, student attempts, automated evaluation, solution generation and performance analysis.

Data Foundation Vaisnavi	AI Intelligence Simpi	Evaluation System Haribabu

Prepared for Team Leader Review

Architecture document • Version 1.0

1. Executive Overview

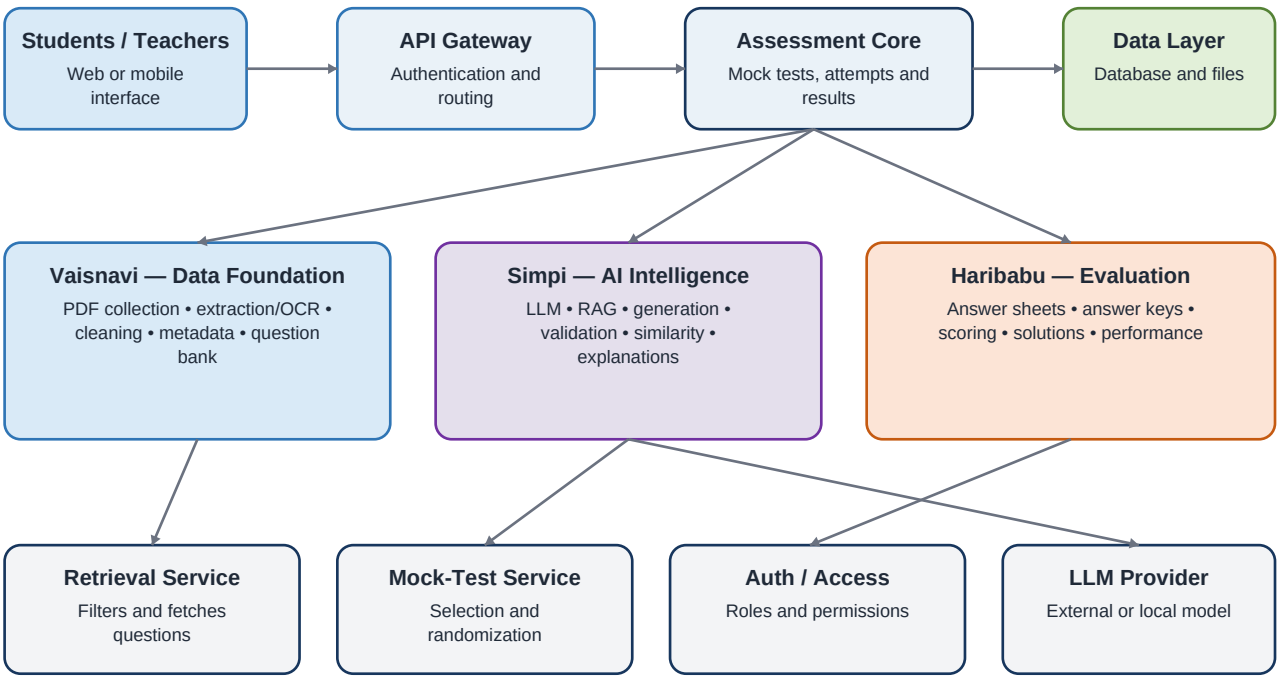
The proposed platform follows a complete assessment lifecycle: **syllabus and question-bank data → mock-test generation → student attempt → answer-sheet processing → evaluation → results → performance analysis**.

The system is organized into three major ownership areas:

Team Member	Primary Ownership	Core Deliverable
Vaisnavi	Question-bank and data pipeline	Clean, structured and searchable question repository with reliable metadata and provenance.
Simpi	LLM and AI intelligence layer	Syllabus-aware generation, RAG, question validation, similarity detection, explanations and AI-assisted subjective evaluation.
Haribabu	Answer-sheet and evaluation system	Answer processing, answer keys, objective/subjective scoring, solutions, results and performance feedback.

Architectural recommendation: begin with a modular monolith. Keep the three ownership areas as separate internal modules with clear interfaces. This reduces deployment complexity while preserving the option to split services later.

2. High-Level System Architecture



Recommended initial implementation: modular monolith with clearly separated module boundaries

The user interface communicates with a central backend/API layer. The backend coordinates the question bank, AI services, mock-test generation, answer processing, evaluation and reporting. Shared services provide authentication, retrieval, storage and model access.

3. Module-Wise Architecture and Responsibilities

3.1 Vaisnavi — Question Extraction, Structuring and Question Bank

Pipeline Stage	Responsibilities	Output
Source collection	Collect board/class/subject/examination question papers and question-bank PDFs.	Source document inventory
Extraction / OCR	Extract text, tables, options, diagrams and page-level content; use OCR where required.	Raw extracted content
Cleaning	Remove noise, fix encoding, normalize spacing and reconstruct question boundaries.	Clean question text
Structuring	Separate question statement, options, marks, answer key, solution and supporting material.	Structured question record
Metadata and validation	Add board, class, subject, chapter, difficulty, question type, marks and source provenance.	Validated metadata
Storage and retrieval	Store records in the central database and expose retrieval APIs for downstream modules.	Searchable question bank

Primary handover to the team: validated question-bank files, database schema, ingestion scripts, metadata dictionary, source-document register and retrieval API documentation.

3.2 Simpi — LLM and AI Intelligence Layer

AI Capability	Responsibilities	Integration Point
Question understanding	Interpret syllabus, class, subject, chapter, difficulty, marks, question type and examination requirements.	Assessment request API
RAG and retrieval	Retrieve relevant question-bank records and syllabus context before generation or explanation.	Retrieval service / database
Question generation	Generate new questions or assemble AI-assisted questions under defined constraints.	Mock-test generation
Validation and QA	Check correctness, syllabus alignment, difficulty, ambiguity, answer consistency and format compliance.	Question approval pipeline
Similarity detection	Detect duplicates and near-duplicates to reduce repetition across tests and the question bank.	Question similarity service
Explanations and hints	Generate understandable explanations, hints and solution rationales where required.	Results and learning support
Subjective evaluation support	Assist with rubric-based analysis, partial marking suggestions and feedback; final scoring rules remain controlled by the evaluation module.	Haribabu evaluation engine

Primary handover to the team: prompt library, model configuration, AI service interface, RAG design, validation rules, similarity strategy, evaluation prompts and sample input/output test cases.

3.3 Haribabu — Answer-Sheet Processing and Evaluation

Evaluation Stage	Responsibilities	Output
Answer submission	Accept student responses through typed answers, uploaded sheets or supported answer formats.	Student attempt record
Answer-sheet processing	Parse, normalize and associate answers with the relevant test questions.	Structured student answers
Answer-key management	Maintain answer keys, model solutions, marking schemes and rubrics.	Evaluation reference set

Evaluation Stage	Responsibilities	Output
Objective evaluation	Evaluate MCQs, true/false, numerical and other objectively scorable responses.	Marks and correctness
Subjective evaluation	Support rubric-based scoring, partial marking and AI-assisted review where appropriate.	Scored response with rationale
Results and analysis	Generate marks, correctness, solutions, weak areas, trends and improvement suggestions.	Result and performance report

Primary handover to the team: answer-processing specification, answer-key format, marking rules, evaluation API, scoring test cases, result schema and performance-analysis logic.

4. End-to-End Assessment Lifecycle

Step	System Activity	Responsible Module
1	Student selects board, class, subject, examination, syllabus area and difficulty.	Frontend / Assessment Core
2	Relevant questions are retrieved from the structured question bank.	Vaisnavi / Retrieval Service
3	AI interprets requirements and supports generation, validation or question enhancement.	Simpi / AI Layer
4	Mock test is assembled, randomized and delivered to the student.	Assessment Core
5	Student submits answers or uploads an answer sheet.	Frontend / Answer API
6	Answers are parsed and mapped to the relevant questions.	Haribabu / Answer Processing
7	Objective and subjective responses are evaluated using keys, rubrics and AI assistance where applicable.	Haribabu + Simpi
8	Marks, correctness, solutions, weak areas and performance insights are generated.	Haribabu / Reporting
9	Results and learning recommendations are displayed to the student or teacher.	Frontend / Reporting

5. Conceptual Data Model

The following entities provide a conceptual starting point for the database. The final schema should be jointly reviewed before implementation.

Entity	Purpose	Important Attributes
SUBJECT / CHAPTER	Organize syllabus and academic hierarchy.	subject_id, class, board, chapter_id, chapter_name
QUESTION	Store normalized question content and metadata.	question_id, text, type, marks, difficulty, source_id
SOURCE_DOCUMENT	Track provenance of extracted questions.	source_id, filename, URL, page, extraction_method
QUESTION_OPTION	Store options for objective questions.	option_id, question_id, option_text, label
ANSWER_KEY	Store correct answers, solutions and marking references.	answer_key_id, question_id, correct_answer, solution, rubric
QUESTION_SIMILARITY	Track duplicate or near-duplicate relationships.	question_a, question_b, similarity_score, decision
MOCK_TEST / TEST_QUESTION	Represent generated tests and their question ordering.	test_id, question_id, sequence, configuration
STUDENT_ATTEMPT / STUDENT_ANSWER	Track a student’s test session and responses.	attempt_id, student_id, answer_text, submitted_at
EVALUATION_RESULT	Store question-level and test-level scoring.	result_id, attempt_id, marks, correctness, feedback
PERFORMANCE_RECORD	Store trends, weak areas and recommendations.	student_id, topic, accuracy, trend, recommendation

6. Proposed API Contracts

These endpoints are proposed interfaces for coordination. Names and payloads should be finalized during backend integration.

Owner	Endpoint	Purpose
Vaisnavi	GET /questions	Retrieve questions using board, class, subject, chapter, difficulty, type or marks filters.
Vaisnavi	GET /questions/{id}	Fetch one complete question record with metadata and provenance.
Vaisnavi	POST /questions	Create or import a validated structured question.
Vaisnavi	GET /subjects	Return available academic hierarchy and syllabus categories.
Simpi	POST /ai/analyze-question	Analyze question quality, structure, syllabus alignment and difficulty.
Simpi	POST /ai/generate-question	Generate a question using supplied constraints and context.
Simpi	POST /ai/validate-question	Validate correctness, ambiguity, format and answer consistency.
Simpi	POST /ai/similarity-check	Check duplicate or near-duplicate questions.
Simpi	POST /ai/explain	Generate a solution explanation, hint or learning feedback.
Simpi	POST /ai/evaluate-subjective	Provide rubric-based AI assistance for subjective answers.
Haribabu	POST /answers/submit	Submit typed answers or answer-sheet references.
Haribabu	POST /evaluation/run	Run objective and/or subjective evaluation.
Haribabu	GET /results/{attempt_id}	Return marks, correctness, solutions and feedback.
Haribabu	GET /solutions/{question_id}	Return model solution and marking reference.
Haribabu	GET /performance/{student_id}	Return topic-wise performance and recommendations.

7. Deployment and Collaboration Strategy

• **Frontend:** web or mobile interface for students and teachers.

• **Backend:** modular monolith containing assessment, question-bank, AI, answer-processing and reporting modules.

• **Database:** relational database for structured questions, tests, attempts, answers, results and metadata.

• **File storage:** separate storage for source PDFs, extracted artifacts, uploaded answer sheets and generated reports.

• **AI runtime:** external LLM provider or local model behind a stable AI service interface.

• **Version control:** shared Git repository with feature branches, pull requests and review before merging.

• **Testing:** unit tests for module logic, integration tests for APIs and validation datasets for AI outputs.

• **Future scalability:** split the AI, extraction or evaluation modules into independent services only when load, ownership or deployment needs justify it.

8. Team Handover Checklist

Team Member	Required Handover Items	Status to Confirm
Vaisnavi	Source-document register; copied PDF/CSV/Excel/JSON artifacts; extraction scripts; cleaned dataset; metadata dictionary; database schema; ingestion instructions; retrieval API documentation; validation report.	Files shared and reviewed
Simpi	Prompt library; model settings; AI module code; RAG workflow; validation rules; similarity-detection method; API schemas; sample prompts and outputs; AI test cases; known limitations.	AI interface tested
Haribabu	Answer-sheet format; answer-key schema; model solutions; marking rules; evaluation engine; scoring test cases; result schema; performance-analysis logic; feedback format.	Evaluation flow tested
All members	README; environment setup; dependency list; configuration template; branch strategy; integration assumptions; open issues; demo instructions.	Team leader sign-off

9. Final Architecture Summary

Vaisnavi	**Assessment Core + Simpi**	**Frontend + Haribabu**	**Haribabu + Simpi**	**Haribabu**

The platform should be developed as one integrated system with explicit module ownership, stable API contracts, shared data definitions and regular integration testing. The immediate priority is to establish a reliable question-bank foundation, a well-defined AI service boundary and a testable evaluation workflow.

Document note: The data model, endpoint names and deployment choices are proposed architecture decisions and should be finalized collaboratively before production implementation.